Exercise session 7

Problem 1

Solve the following non-linear equality-constrained problem

$$\begin{aligned} \min \quad & x_3 \\ \text{s.t.} \quad & x_1^2 + x_2^2 + x_3^2 = 6 \\ & x_1 + x_2 + x_3 = 4. \end{aligned} \tag{NLP$_=$}$$

Motivate global optimality.
Hint: Show that $\mathcal{F}$ is compact.

Problem 2

Solve the following non-linear inequality-constrained problem

$$\begin{aligned} \min \quad & cx_1 + x_2 \\ \text{s.t.} \quad & (x_1 - 1)^2 + (x_2 - 1)^2 \leq 9 \\ & (x_1 - 1)^2 + (x_2 + 1)^2 \leq 9 \\ & (x_1 + 1)^2 + (x_2 - 1)^2 \leq 9 \\ & (x_1 + 1)^2 + (x_2 + 1)^2 \leq 9 \end{aligned} \tag{NLP$_\leq$}$$

for the values $c = 0$ and $c = 1$.
Motivate global optimality.

Problem 3

Likely not treated during class, you can work with it yourself.

Consider the following non-linear inequality-constrained problem

$$\begin{aligned} \min \quad & x_1 \\ \text{s.t.} \quad & x_2 - x_1^3 \leq 0 \\ & -x_2 - x_1^3 \leq 0. \end{aligned} \tag{$\widehat{\text{NLP}}_\leq$}$$

a) Illustrate the feasible region and draw level curves of the objective function. Use this to show that $\hat{x} = (0, 0)^T$ is a global minimizer.

b) Show that $\hat{x}$ does not satisfy the KKT-conditions for ($\widehat{\text{NLP}}_\leq$). Explain why.